

Abstract

This project develops a digital AC energy meter using Arduino Uno, ACS712 current sensor, transformer-based voltage sensing circuit and LCD display. The system measures voltage, current, power and energy in real time using Proteus simulation.

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Introduction

Electrical energy measurement is essential in residential, commercial and industrial systems. Digital energy meters provide real-time monitoring and improved accuracy. Arduino-based solutions offer a low-cost platform for learning and development.

Literature Review

Traditional electromechanical meters have been replaced by digital energy meters. Modern designs use microcontrollers, sensors and digital displays to monitor electrical parameters. Arduino-based systems are popular in educational and research applications.

Objectives

Measure AC Voltage, AC Current, Power Consumption and Energy Consumption. Display parameters on LCD and validate operation through Proteus simulation.

Hardware Requirements

1	Arduino Uno
2	ACS712-05B Current Sensor
3	20x4 LCD
4	220V/13V Transformer
5	100k Ω and 10k Ω Resistors
6	10 μ F Capacitor

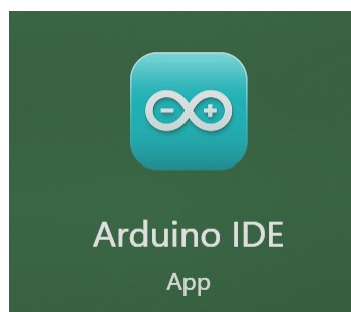


Fig: Arduino IDE

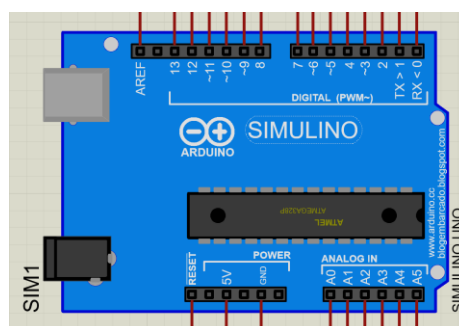


Fig: Arduino UNO

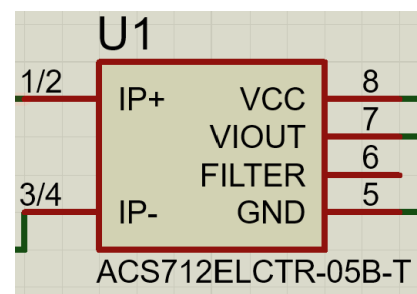


Fig: Current Sensor

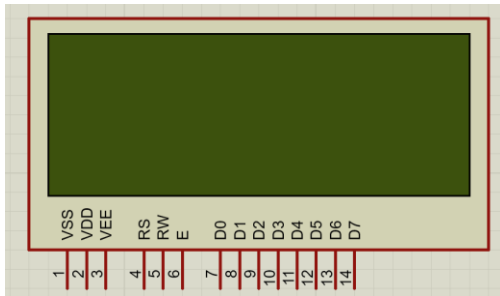


Fig: LCD 20x4

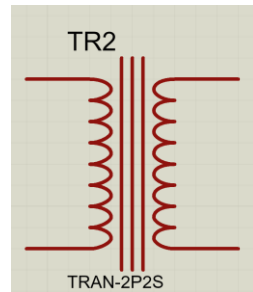


Fig: Transformer

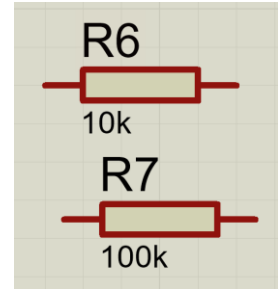


Fig: Resistors

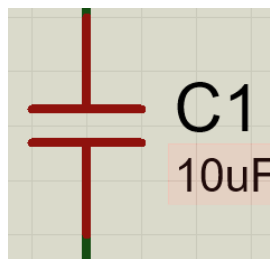


Fig: Capacitor

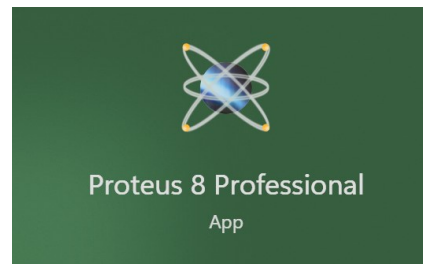


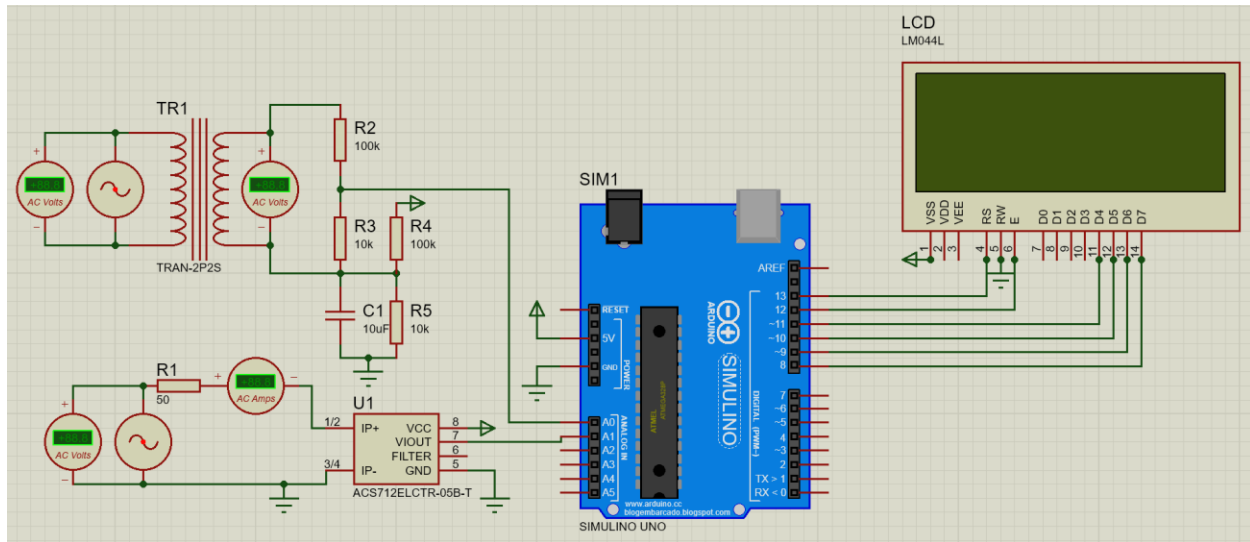
Fig: Proteus 8 Professional

Software Requirements

Arduino IDE, EmonLib Library, Proteus Design Suite 8 Professional.

Circuit Description

The transformer steps down 220V AC to approximately 13V AC. The resistor divider and capacitor network condition the signal for Arduino ADC measurement. ACS712 measures current. Arduino computes electrical parameters and displays them on LCD.



Working Principle

Voltage is sensed through a transformer and divider network. Current is sensed using ACS712. Arduino acquires analog data, calculates RMS voltage, current, power and energy, and updates the LCD continuously.

Mathematical Calculations

Voltage Measurement: V_{rms} obtained using EmonLib.

Current Measurement: $I = (\text{Sensor Voltage} - \text{Offset}) / \text{Sensitivity}$.

Power: $P = V \times I$.

Energy: $E = (\text{Power} \times \text{Time}) / 3600 \text{ Wh}$.

Algorithm

1. Initialize LCD.
2. Initialize EmonLib.
3. Read voltage.
4. Read current.
5. Calculate power.
6. Calculate energy.
7. Display values.
8. Repeat.

Program Code Explanation

The Arduino program uses LiquidCrystal library for LCD interfacing and EmonLib for RMS voltage measurement. ACS712 output is processed through analog acquisition and calibration.

Program Code

```
// include the library code:
```

```
#include <LiquidCrystal.h> //library for LCD
```

```
// initialize the library with the numbers of the interface pins
```

```
LiquidCrystal lcd(13, 12, 11, 10, 9, 8);
```

```
// Measuring Voltage
```

```
#include "EmonLib.h" // Include Emon Library
```

```
EnergyMonitor emon1; // Create an instance
```

```
// Measuring Current Using ACS712
```

```
const int Sensor_Pin = A1; //Connect current sensor with A1 of Arduino
```

```
int sensitivity = 185; // use 100 for 20A Module and 66 for 30A Module
```

```
int offsetvoltage = 2542;
```

```
void setup()
```

```
{
```

```
  emon1.voltage(A0,267, 1.7); // Voltage: input pin, calibration, phase_shift
```

```
  lcd.begin(20, 4); // set up the LCD's number of columns and rows:
```

```
  lcd.setCursor(0,0);
```

```
  lcd.print(" AC ENERGY METER      ");
```

```
}
```

```
void loop()
```

```
{
```

```
  //***** Measure Voltage *****
```

```
emon1.calcVI(20,2000); // Calculate all. No.of half wavelengths (crossings), time-out
```

```
int Voltage = emon1.Vrms; //extract Vrms into Variable
```

```
  lcd.setCursor(0,1);
```

```
  lcd.print("V = ");
```

```
  lcd.print(Voltage);
```

```
  lcd.print("V ");
```

```
//***** Measure Current *****
```

```
unsigned int temp=0;
```

```
float maxpoint = 0;
```

```
for(int i=0;i<500;i++)
```

```
{
```

```
  if(temp = analogRead(Sensor_Pin), temp>maxpoint)
```

```
  {
```

```
    maxpoint = temp;
```

```
  }
```

```
}
```

```
float ADCvalue = maxpoint;
```

```
double eVoltage = (ADCvalue / 1024.0) * 5000; // Gets you mV
```

```
double Current = ((eVoltage - offsetvoltage) / sensitivity);
```

```
double AC_Current = ( Current ) / ( sqrt(2) );
```

```
lcd.print("I = ");
```

```
lcd.print(AC_Current,1);
```

```
lcd.print("A      "); //unit for the current to be measured
```

```
//***** Measure Power *****
```

```

float Power = Voltage * AC_Current;

lcd.setCursor(0,2);

lcd.print("P = ");

lcd.print(Power);

lcd.print("W "); //unit for the current to be measured

//***** Measure Energy *****

long milisec = millis(); // calculate time in milliseconds

long time=milisec/1000; // convert milliseconds to seconds

float Energy = (Power*time)/3600; //Watt-sec is again convert to Watt-Hr by dividing
1hr(3600sec)

lcd.print("E = ");

lcd.print(Energy,0);

lcd.print("Wh "); //unit for the current to be measured

delay(200);

}

```

Simulation Results

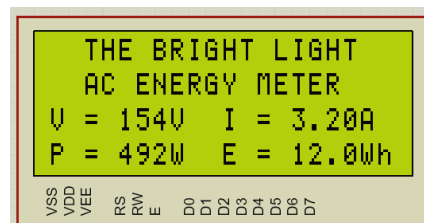
Input Voltage = 220V

Transformer Output = 13V

Measured Current $\approx 3.2\text{A}$

Measured Power $\approx 703.74\text{W}$

Energy displayed in Wh. The simulation operated successfully and produced stable readings.



Result Analysis

Measured values closely matched theoretical values. The system successfully demonstrated voltage sensing, current sensing, power calculation and energy monitoring.

Advantages

Low Cost, Easy Implementation, Real-Time Monitoring, Educational Value, Flexible Expansion.

Limitations

Power factor assumed unity, simplified energy calculation, simulation-oriented implementation.

Applications

Home Energy Monitoring, Smart Metering, Educational Laboratories, Electrical Experiments, Power Monitoring Systems.

Future Scope

IoT Integration, GSM Alerts, Cloud Monitoring, Data Logging, Mobile Applications and Smart Grid Integration.

Conclusion

The project successfully demonstrates an Arduino-based AC energy meter capable of measuring voltage, current, power and energy in Proteus simulation.

IEEE References

- [1] Arduino Uno Datasheet.
- [2] ACS712 Datasheet.
- [3] OpenEnergyMonitor EmonLib Documentation.
- [4] Proteus Design Suite Documentation.
- [5] JNTUK Laboratory Manuals.